

Kubernetes: A Dependable and Popular Platform

Kubernetes is more than just a tool; it serves as a robust platform, streamlining the deployment of applications, as well as their scaling and operation in various environments.

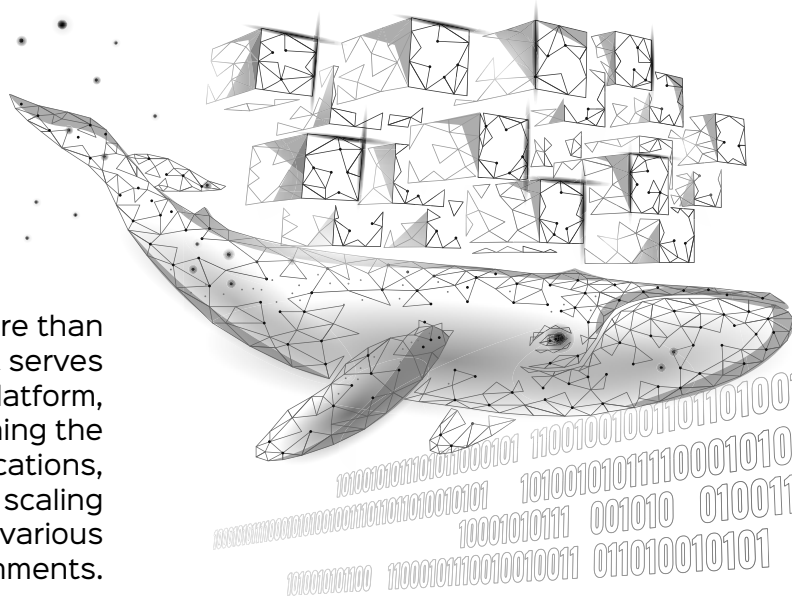


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Kubernetes, often referred to as 'K8s' or 'kube', is an open source container orchestration platform. Its primary role is to automate software scaling, management, and deployment.

Kubernetes is a Greek name that means helmsman or pilot. K8s, the abbreviation, indicates the eight letters between the 'K' and the 's'. Google open sourced the Kubernetes project in 2014. Kubernetes combines over 15 years of Google's experience running production workloads at scale with best-of-breed ideas and practices from the community. It was donated to the Cloud Native Computing Foundation (CNCF) in 2015.

What is Kubernetes?

Kubernetes automates operational tasks of container management and includes built-in commands for deploying applications, scaling applications up and down to fit changing needs, monitoring applications, and making it easier to manage them.

▪ **Kubernetes architecture and operations**

Containers serve as a portable and easily deployable form for encapsulating applications. Kubernetes, designed specifically for running containerised applications, has a distinctive architecture. In a Kubernetes cluster, there is a minimum of one control plane and one worker node (often a physical or virtual server). The control plane has dual responsibilities: exposing the Kubernetes API through the API server and overseeing the cluster's constituent nodes. It plays a key role in decision-making regarding cluster management and effectively responds to various cluster events.

At the core of Kubernetes is the concept of a pod, which represents the smallest execution unit for an application. Pods, comprising one or more containers, are executed on worker nodes.

▪ **Key components of the Kubernetes control plane and nodes**

The control plane of Kubernetes

comprises four essential components responsible for managing communications, node administration, and maintaining the cluster's state.

- **Kube-apiserver:** As implied by its name, kube-apiserver serves the role of exposing the Kubernetes API.
- **etcd:** This is a key-value store housing all data about the Kubernetes cluster, ensuring efficient data management.
- **Kube-scheduler:** Tasked with monitoring new Kubernetes pods lacking assigned nodes, kube-scheduler allocates them to a suitable node based on considerations such as resources, policies, and 'affinity' specifications.
- **Kube-controller-manager:** All functions related to controllers within the control plane are consolidated into a unified binary known as kube-controller-manager. A Kubernetes node is equipped with three major components.